

JUNE 19, 2026

TEAM 12

INTERMEDIATE PRESENTATION

██████████ Abdullah Al Ayaat

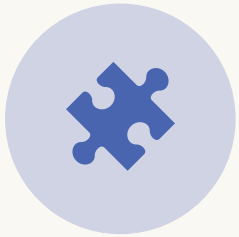
██████████ G. C. Mohnish

██████████ BURTENSHAW Catherine Ella

██████████ TAN Daphne Jillian Gan

██████████ ZAHID Muhammad Abdullah

PRESENTATION AGENDA



ROBOT
DEVELOPMENT



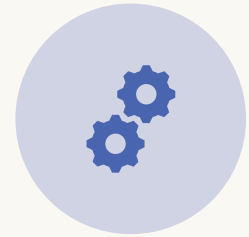
PROBLEMS &
SOLUTIONS



STAGE I & II
APPROACH



MEMBER
CONTRIBUTION



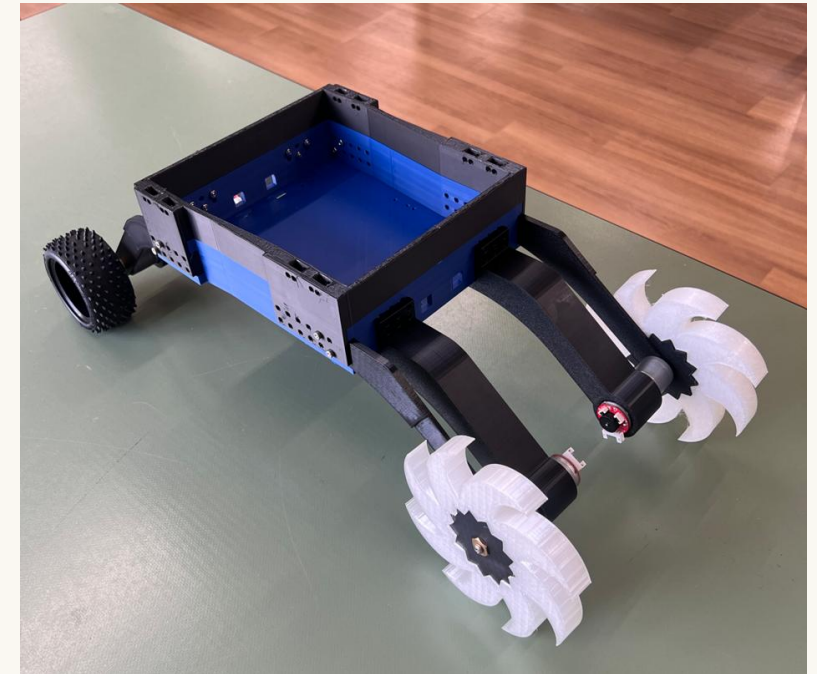
TIMELINE OF
ROBOT
DEVELOPMENT

Robot Development

Tri-Wheel Robot



Four-Wheel Robot



Stability issue due to tri-wheel design

ROBOT DEVELOPMENT (BODY: DESIGN)

Chassis

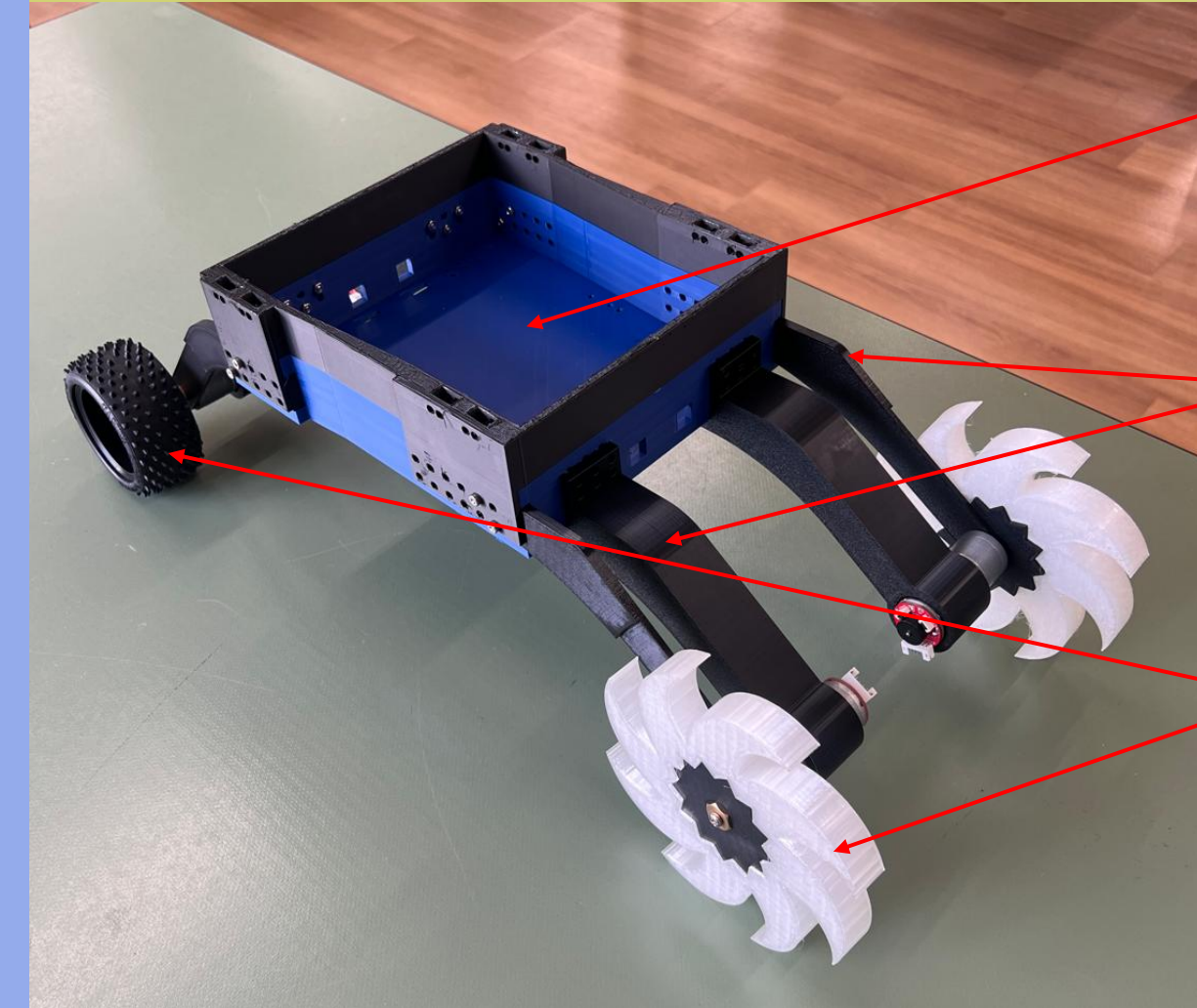
- **Fourth Draft** completed and printed
- Redesigned for improved component integration

Brackets

- Front and rear wheel brackets designed and printed
- Fit and mounting tests performed

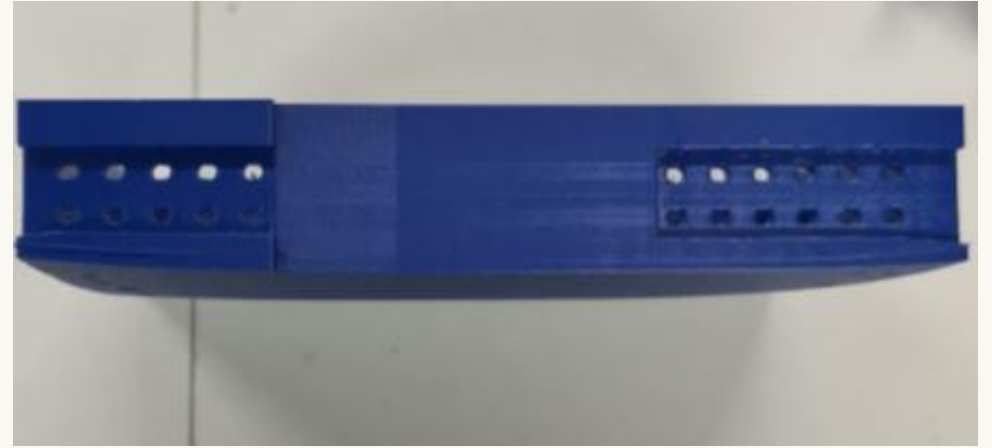
Wheels + Attachments

- TPU front wheel prototypes printed
- Flexible wheel design developed
- Attachment compatibility verified

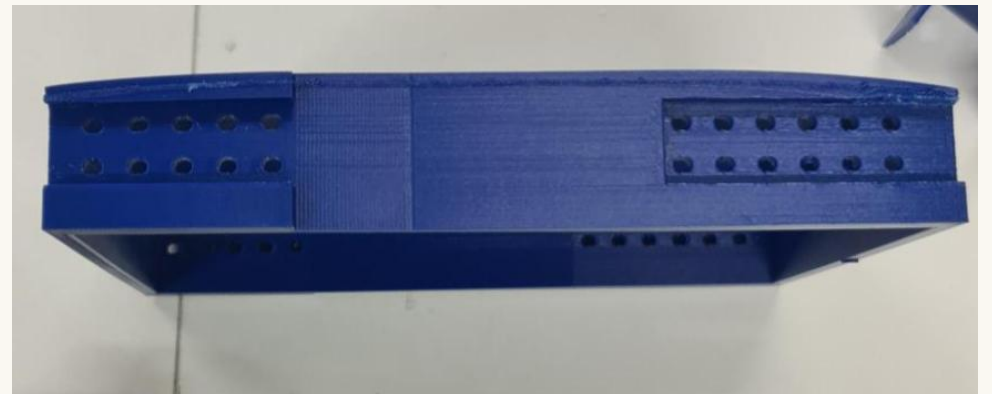


Robot Development : Chassis

- Early drafts suffered from a **base that was too thin**.
- The **second draft** had **dimensions too small** to accommodate **all components**
- The **third draft** experienced **warping** due to incorrect slicer settings and **poor bed adhesion**.

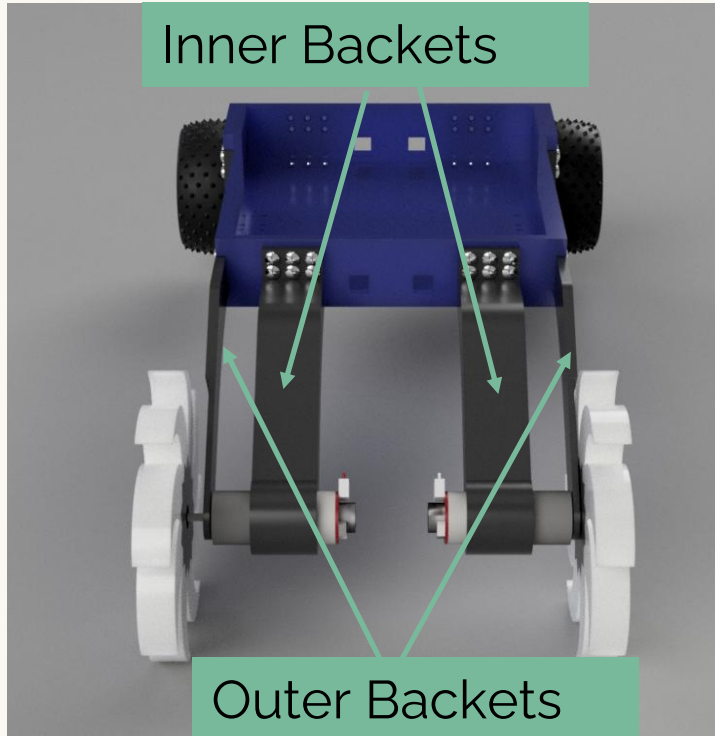


3rd Draft

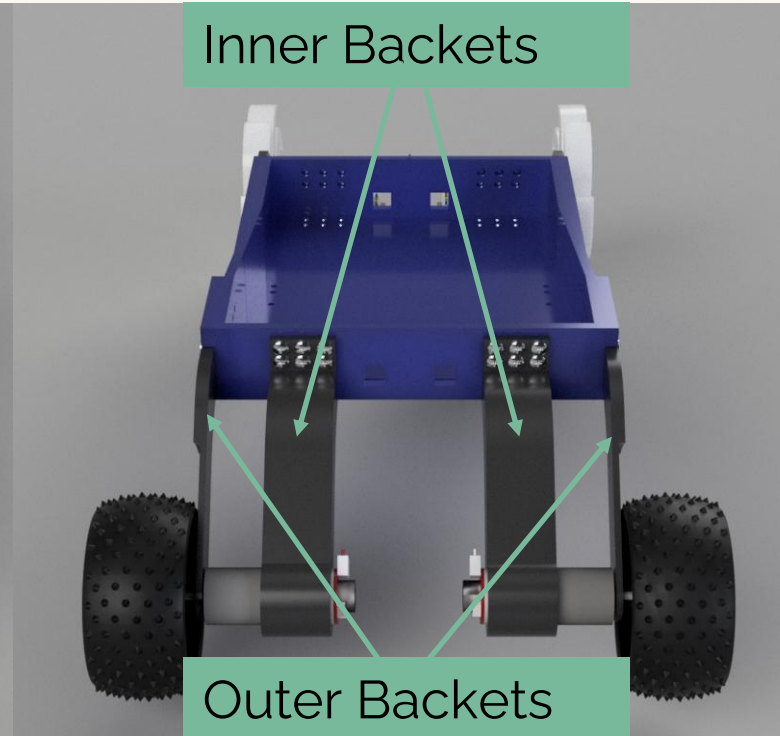


Robot Development : Brackets

Front view:



Back view:



- **Inner Brackets:**
Hold the motors in place
- **Outer Brackets:**
Supports the wheels and rotating motor shafts

Robot Development :Wheels + Attachments

Front Wheels

Inner wheel:

- Strategic holes to **tighten** brass connector from **outside the wheel**

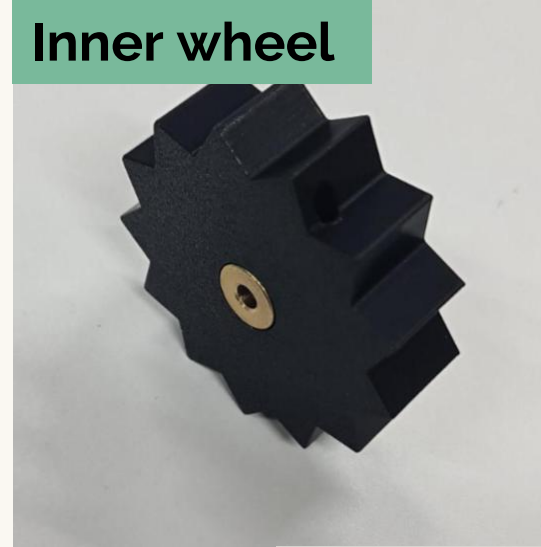
Outer wheel:

- Made with **TPU**: moderately flexible + some shock absorption
- Spiked design = **grip ledges** (stairs)

Back Wheels

- **Spiked** rubber grips

Inner wheel



Outer wheel



Back wheel



ROBOT PROGRESS (BODY: MECHANICAL)

ISSUES & SOLUTIONS

Chassis Warping

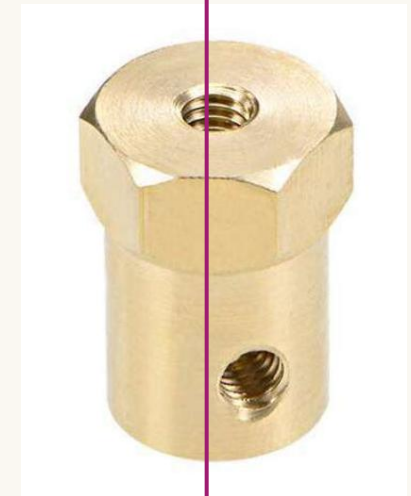
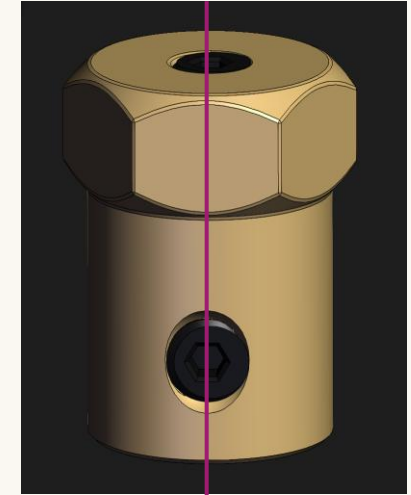
- Consulted TAs regarding print parameters
- Adjusted print settings
- Successfully reprinted chassis without warping

Hole Alignment (Chassis)

- Revised chassis CAD design
- Standardized screw size (3mm)

Hole Alignment (Wheel Attachment)

- Found couplings that matched the 3D Model



Robot Development (internal components)



LIPO BATTERY 12V



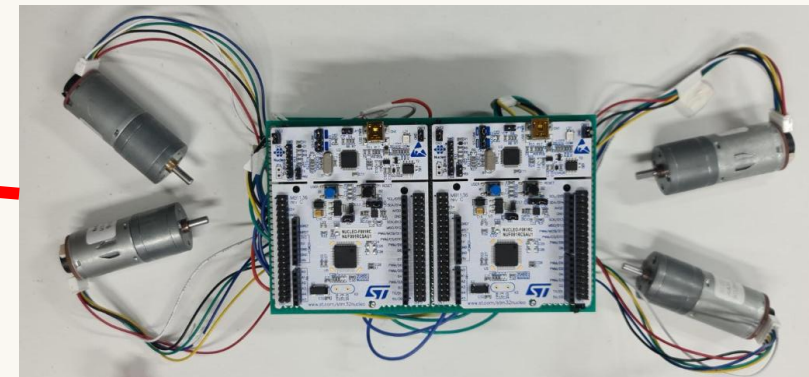
Power Bank



Raspberry pi 4



Buck Converter



Robot Development (internal components)

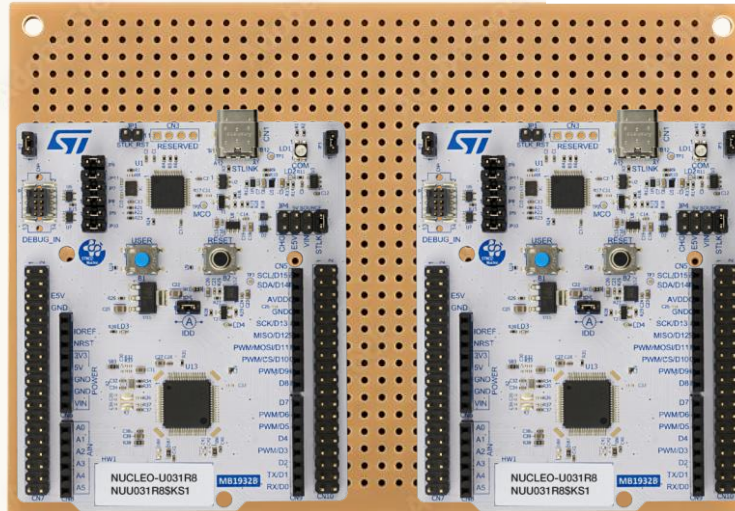
LIPO BATTERY **12V**

Power Bank



SERVO
CIRCUIT

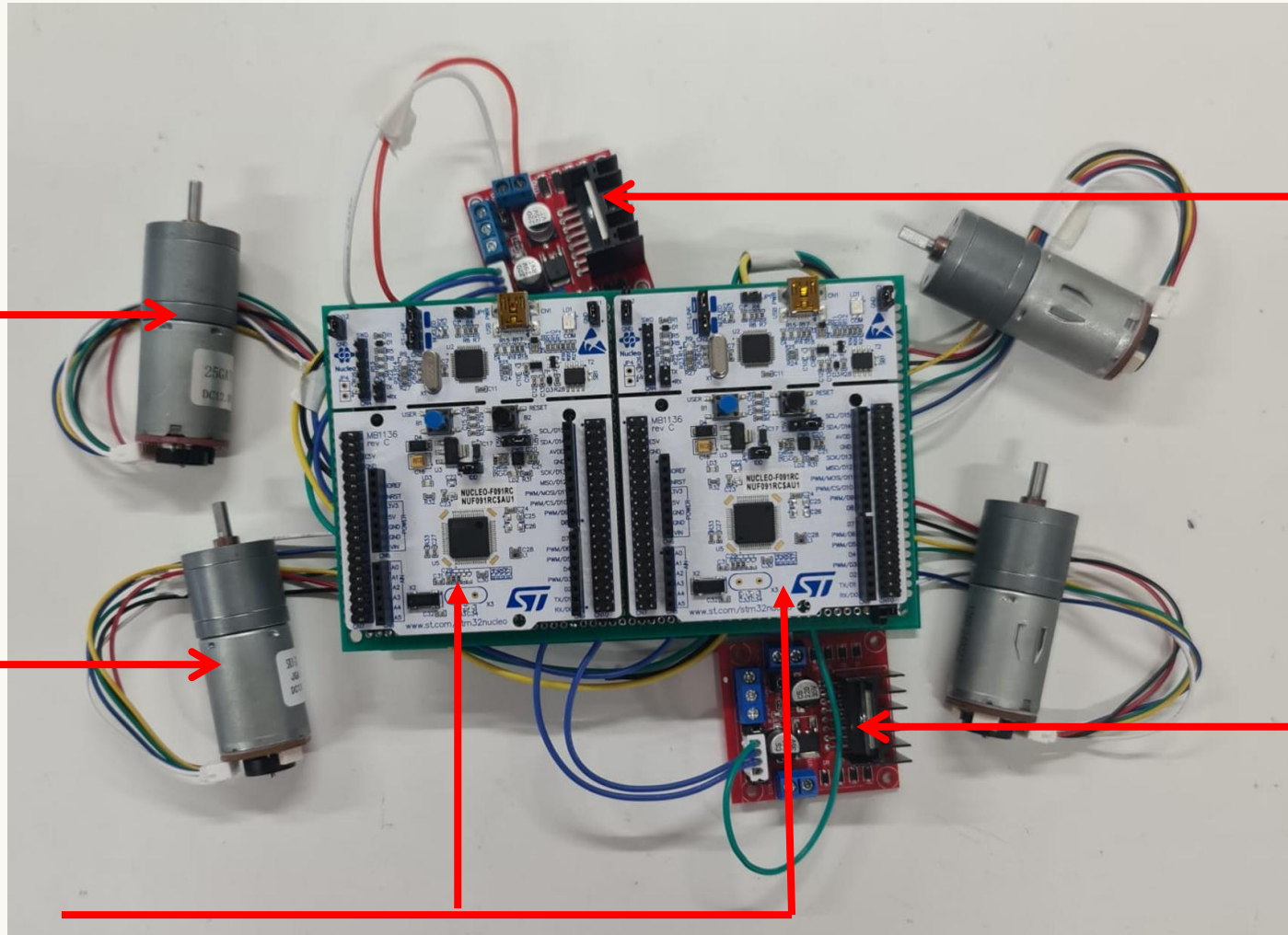
POWER CIRCUIT



ROBOT PROGRESS (GEARED MOTOR CIRCUIT)

4-Geared DC Motor

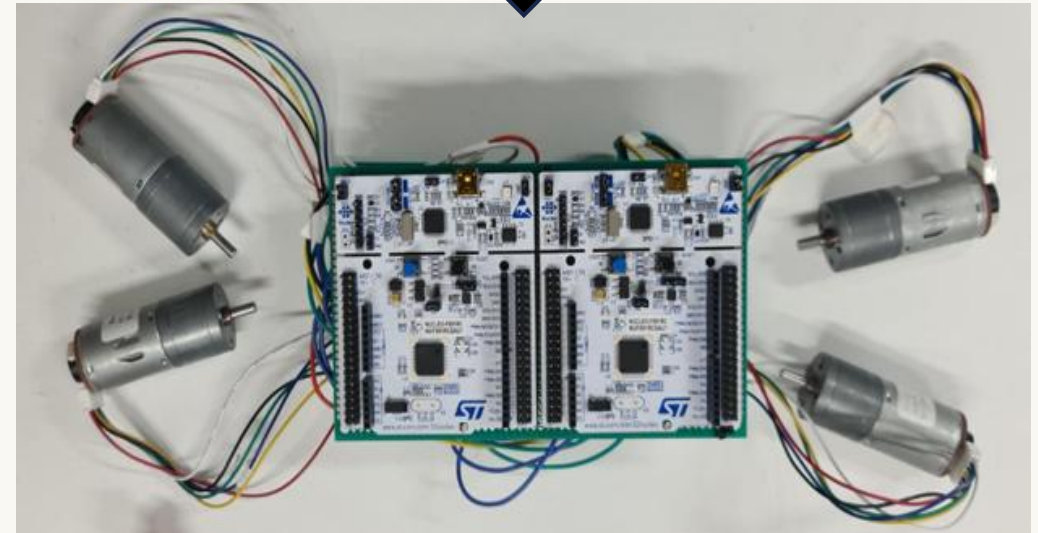
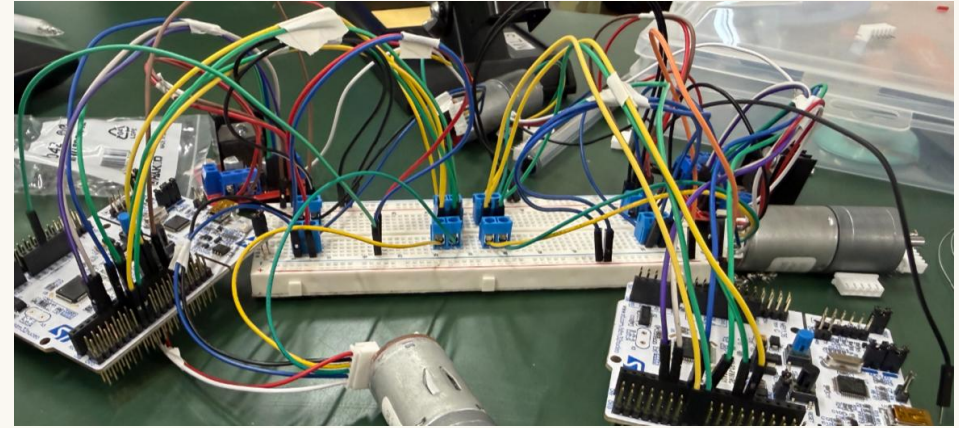
2 -L298N Motor Drivers



2-STM32 Microcontroller

ROBOT PROGRESS (GEARED MOTOR CIRCUIT)

- We built a **test breadboard circuit** to write and test our code, as well as **validate our design**
- We moved to a **perf-board** for stronger connections and a more **compact design** to fit inside the robot



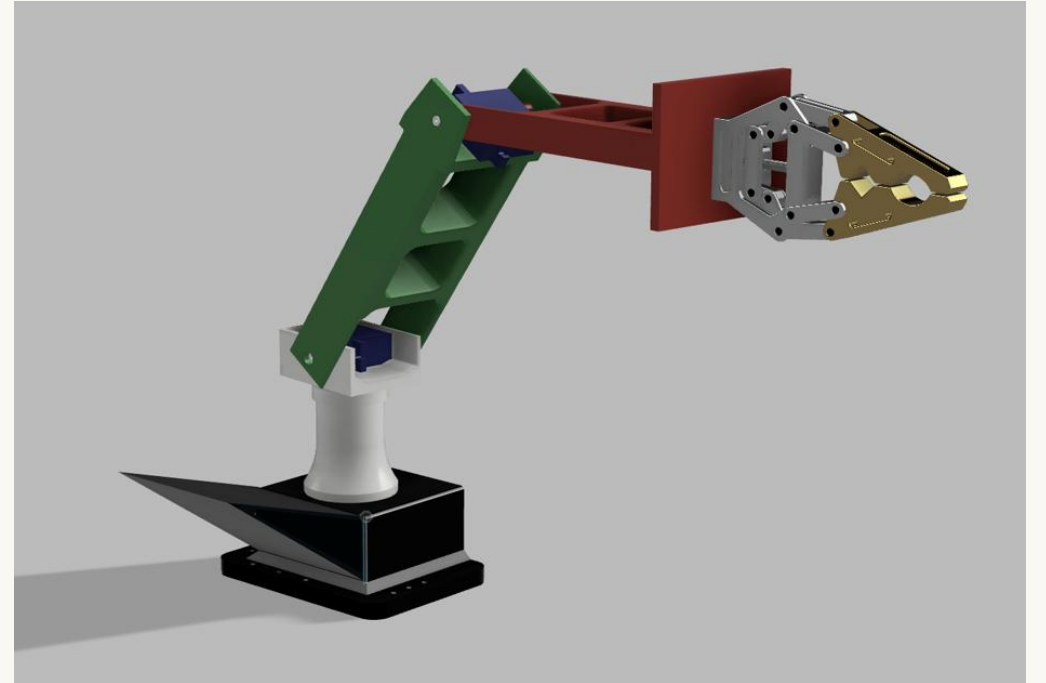
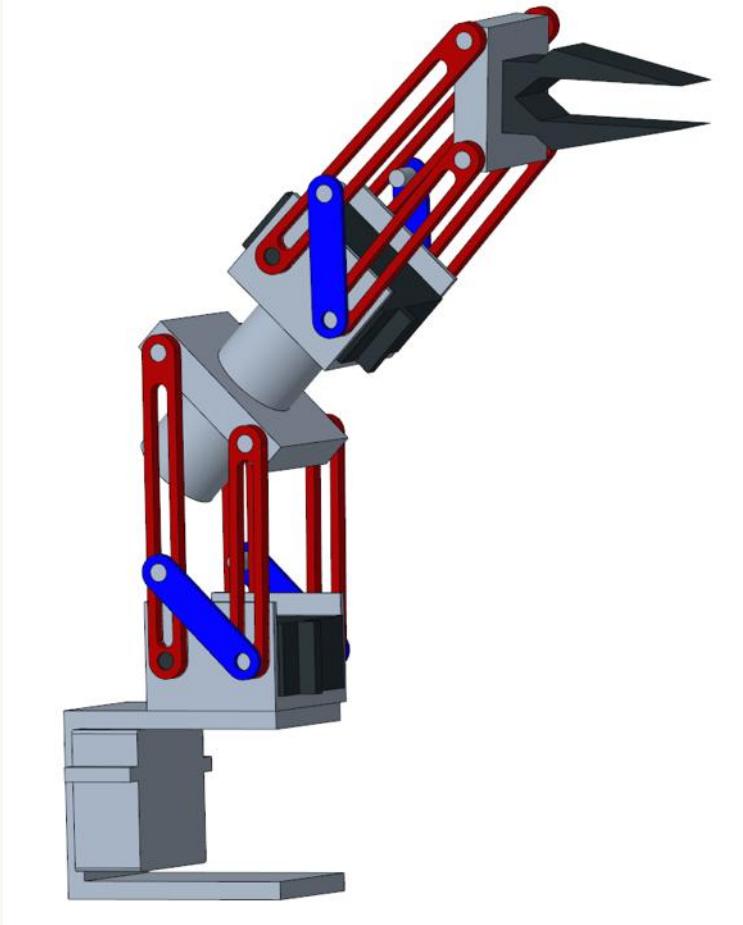
ROBOT PROGRESS (BODY: CODING)

Accomplished:

- Communication between server and raspberry pi.
- Grounding Dino on the server for open-vocabulary object detection – allowing detection of:
 - The doorknob
 - QR Code Cardboard
 - Red Cube Object
- Program for black line detection and motion command JSONs.
- Program for controlling four motors independently.



ROBOT PROGRESS (MANIPULATOR DESIGN)



ROBOT PROGRESS (MANIPULATOR: DESIGN)

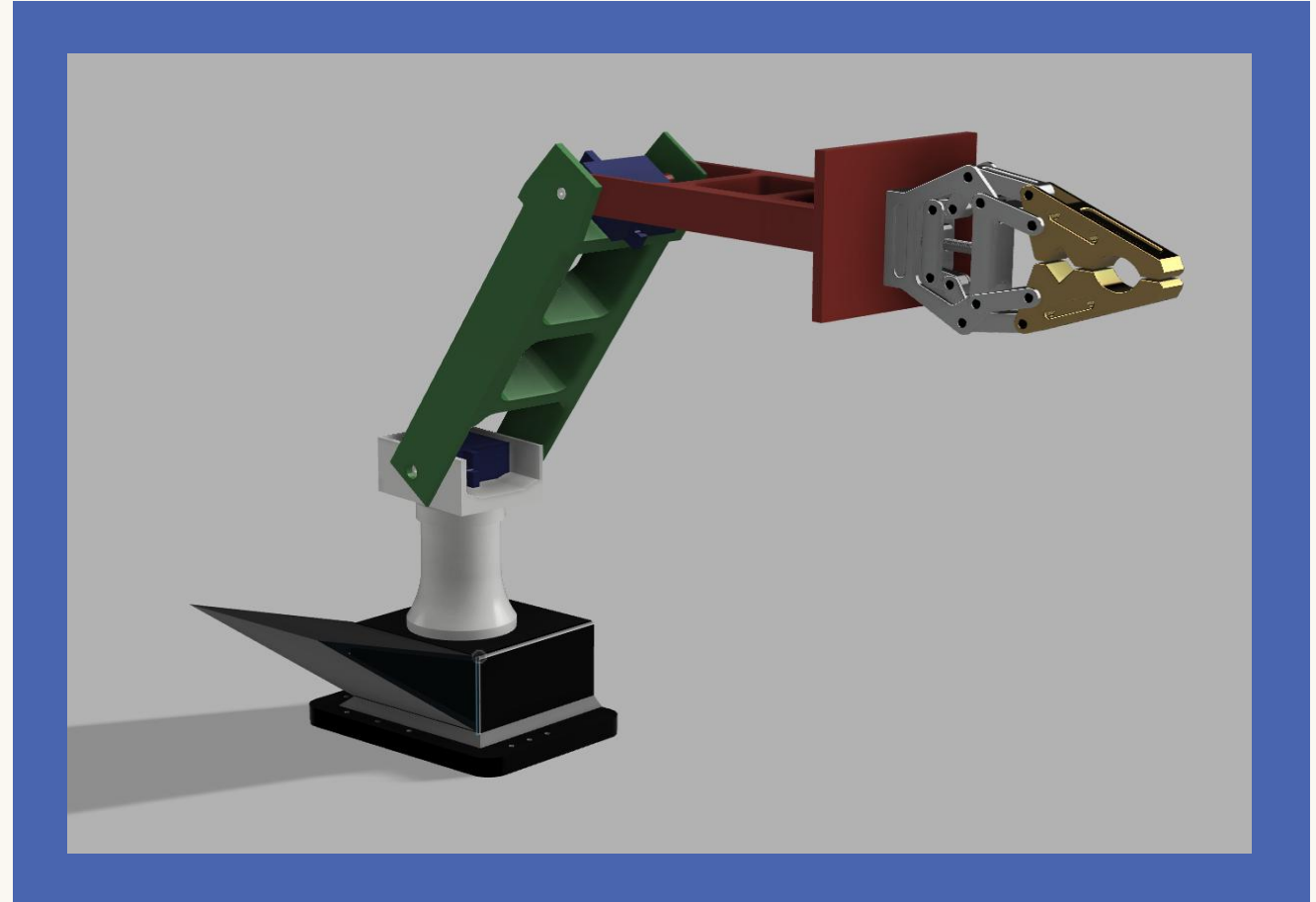
FOCUS ON WEIGHT
DISTRIBUTION

REDUCTION OF
MOTORS

WEIGHT DISTRIBUTION

EASIER WIRE AND
MOTOR PLACEMENT

UPDATED METHOD FOR
RESTING POSITION



ROBOT PROGRESS (MANIPULATOR)

- The 3R-manipulator consists of 3 servo motors that provide us with **3DoF**
- With the help of a **solenoid actuator**, we can open and close the pincher of the manipulator



CURRENT PROBLEMS(+ POSSIBLE SOLUTIONS)

- **Large Robot Footprint/Insufficient Torque**

Limits turning capability in confined areas.

We require stronger motors for overcoming the stairs and turning

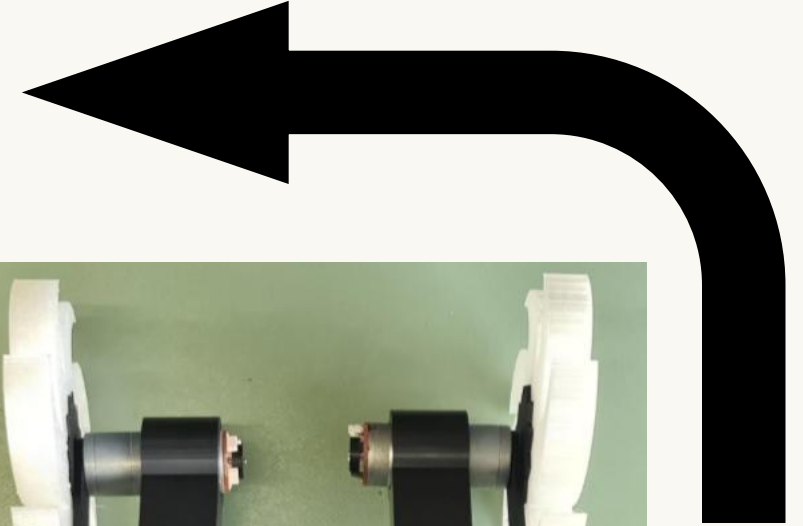
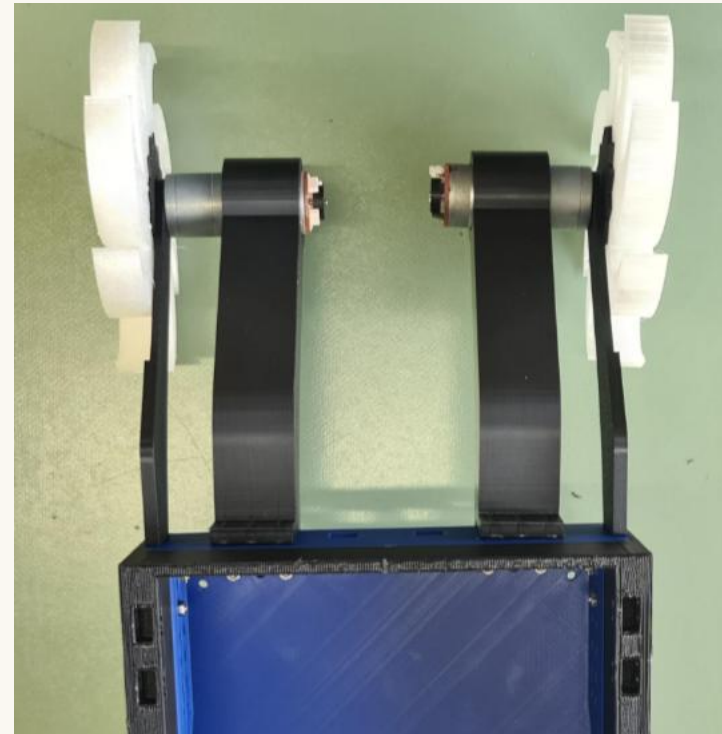
- **Solution:**

Reduce central compartment length.

Stack electronics vertically.

Move brackets beneath the chassis to decrease footprint.

Calculate the exact torque required due to design change



CURRENT PROBLEMS(+ POSSIBLE SOLUTIONS)

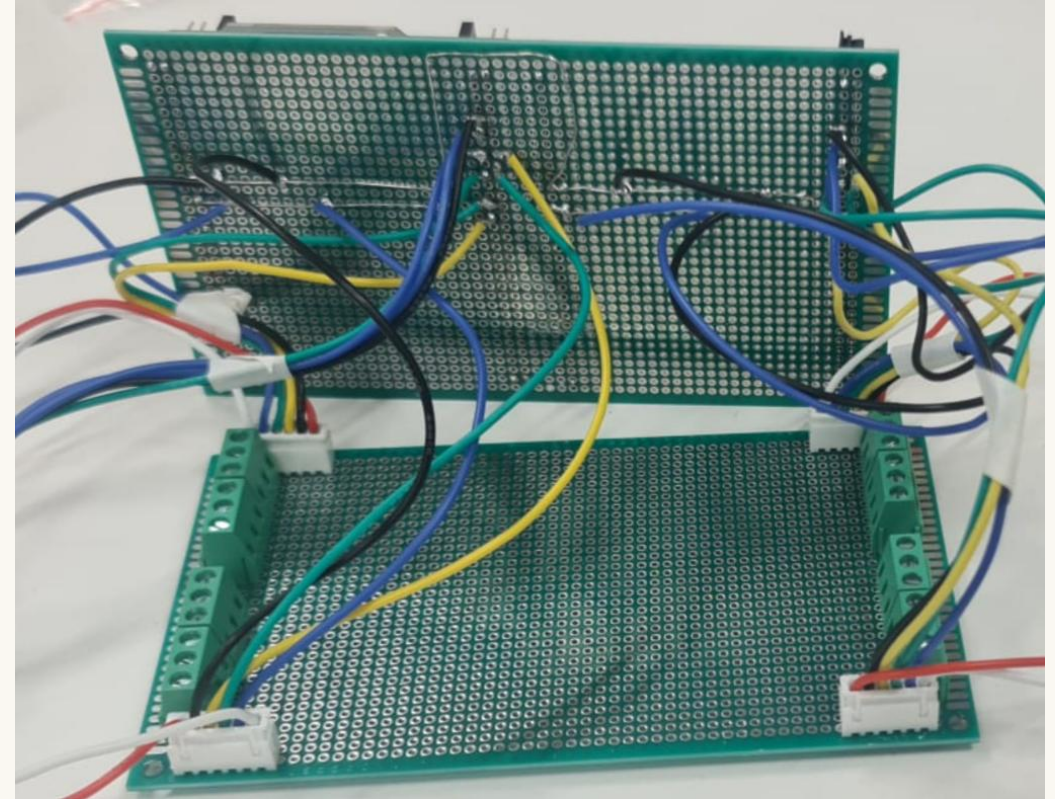
- **4-Motor Control Integration**

Difficulty controlling four motors with a single or double microcontroller.

- **Solution:**

Continue hardware and software debugging.

Validate motor communication and control reliability.



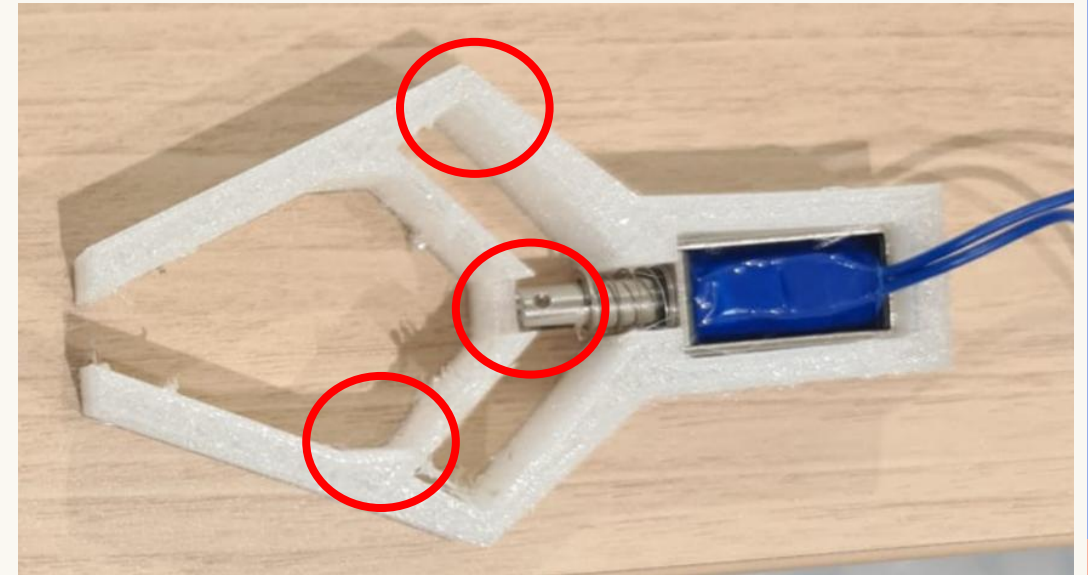
CURRENT PROBLEMS(+ POSSIBLE SOLUTIONS)

- **Manipulator Pincher**

The pincher mechanism proved too stiff or rigid for the solenoid actuator to produce sufficient force to engage and grip the door

- **Solution:**

Incorporate a rigid-flexible hybrid design with **compliant hinges** to reduce pincher stiffness and improve actuation.



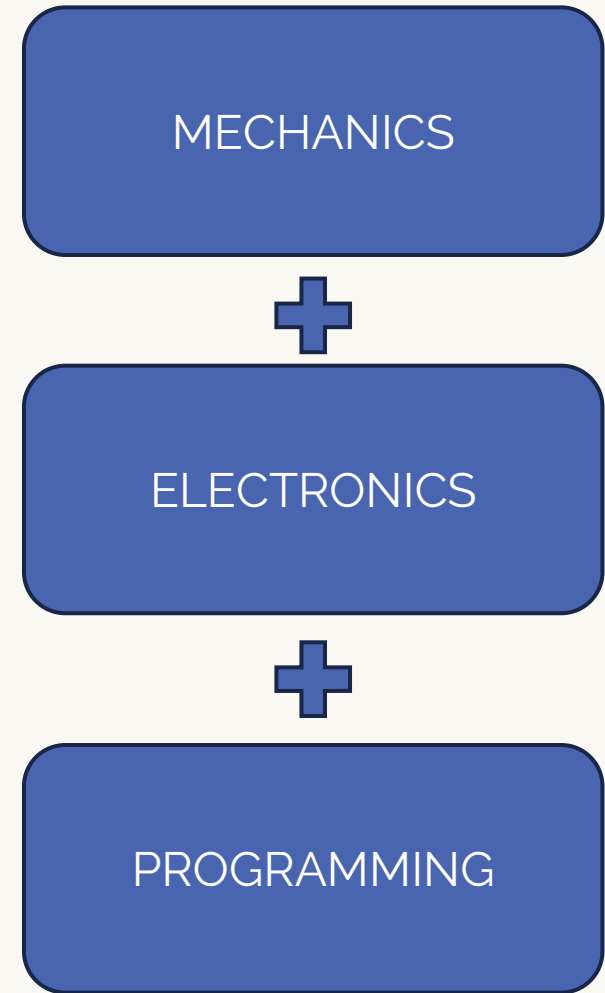
CURRENT PROBLEMS(+ POSSIBLE SOLUTIONS)

- **System Integration & Tuning**

Individual modules are developed but not fully tested together.

- **Next Steps:**

Integrate all subsystems.
Fine-tune control algorithms on the complete robot.



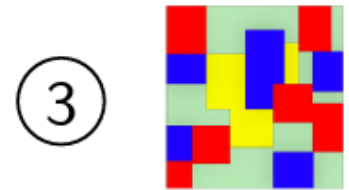
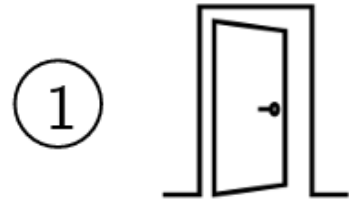
Approach for Stage-I

The robot will push the door

Robot design with back wheel

3D printed claw wheel

Scanning using camera



Robot design with back wheel

3D printed claw wheel

Using camera

2R manipulator with Solenoid-Driven Pincher

⑥



⑦



⑧



⑨

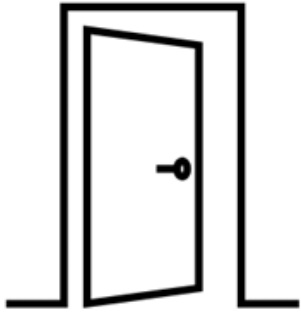


⑩



Approach for Stage-II

①



The door must be
opened not pushed

⑤



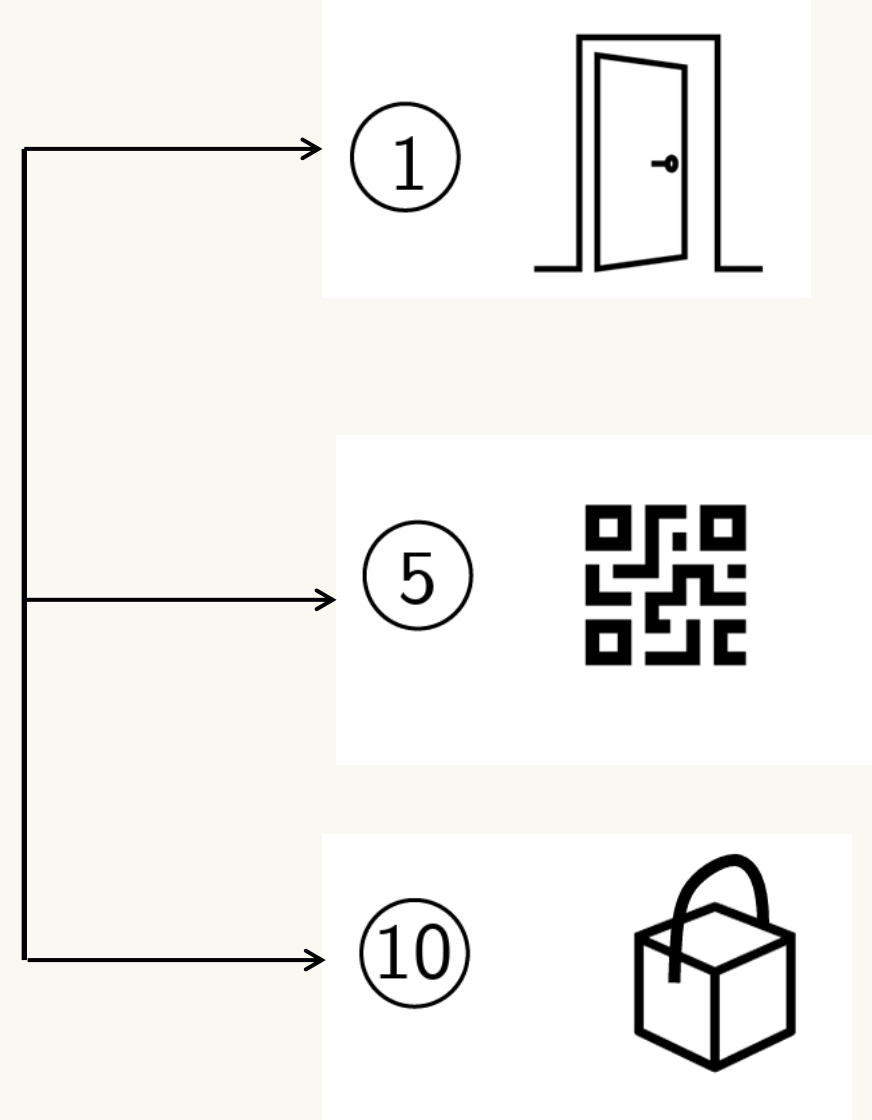
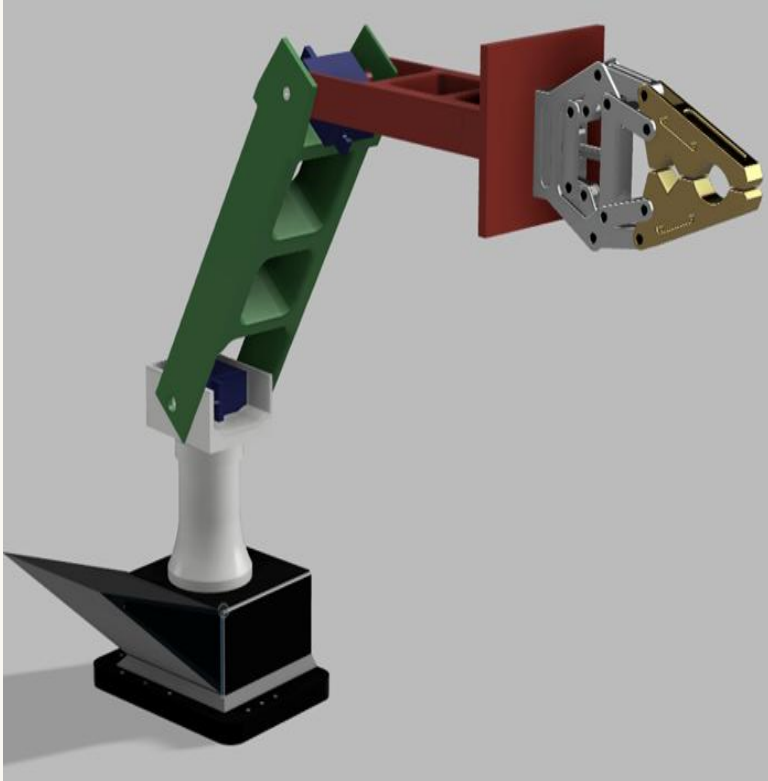
The QR code
must be
flipped

⑩



The Block must
be **picked**

Approach for Stage-II



MEMBER CONTRIBUTION

Ayaat :

- Main Body concept design
- Front Bracket initial Design
- Perf-board soldering
- Designing and Completing the final manipulator arm

Mohnish :

- Initial manipulator arm design
- Initial manipulator 3D Printing
- Electrical circuit design
- Circuit assembly and testing

Abdullah :

- Setting up Grounding Dino
- Developing line detection algorithm
- Setup up key based control for robot motors

Cat :

- Main Robot Body CAD Design
- Wheel (+ attachments) Design
- MWS 3D Printing
- Recent Bracket CAD Design

Daphne :

- Rear bracket initial design
- Component research
- MWS 3D Printing
- Ordering & machine booking

PROJECT TIMELINE

	WEEK																
TASK	7		8		9		10		11		12		13		14		15
	5/26	5/29	6/2	6/5	6/9	6/12	6/16	6/19	6/23	6/26	6/30	7/3	7/7	7/10	7/14	7/17	7/24
Design development																	
Parts manufacturing																	
Prototyping																	
Further development																	
Preparation and test runs																	
Presentation																	

THANK YOU :)

Q&A